



2.3 VARIABLE CAPACITORS

MIKROELEKTRONIKA

Variable capacitors are capacitors with variable capacity. Their minimal capacity ranges from 1p and their maximum capacity goes as high as few hundred pF (500pF max). Variable capacitors are manufactured in various shapes and sizes, but common features for them is a set of fixed plates (called the stator) and a set of movable plates. These plates are fitted into each other and can be taken into and out of mesh by rotating a shaft. The insulator (dielectric) between the plates is air or a thin layer of plastic, hence the name variable capacitor. When adjusting these capacitors, it is important that the plates do not touch.

Below are photos of air-dielectric capacitors as well as mylar-insulated variable capacitors (2.5a).

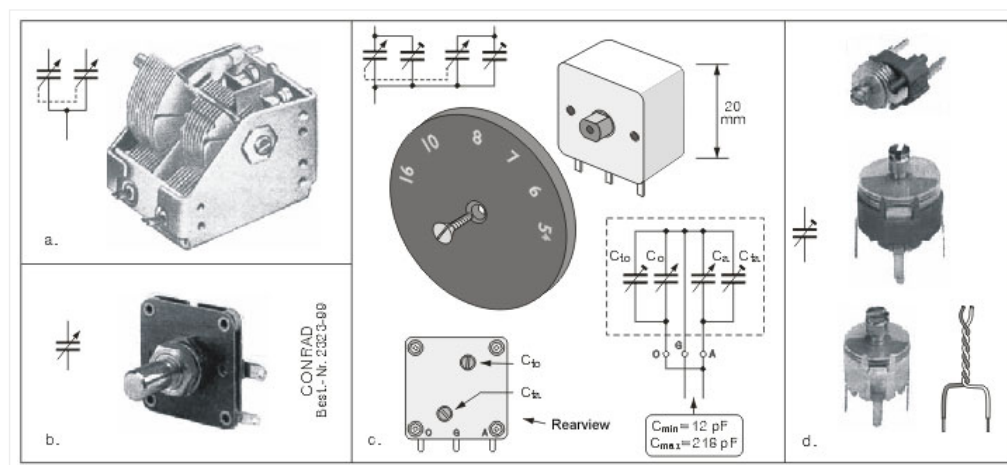


Fig. 2.5: a, b, c. Variable capacitors, d. Trimmer capacitors

The first photo shows a “ganged capacitor” in which two capacitors are rotated at the same time. This type of capacitor is used in radio receivers. The larger is used for the tuning circuit, and the smaller one in the local oscillator. The symbol for these capacitors is also shown in the photo.

Beside capacitors with air dielectric, there are also variable capacitors with solid insulator. With these, thin insulating material such as mylar occupies the space between stator and rotor. These capacitors are much more resistant to mechanical damage. They are shown in figure 2.5b.

The most common devices containing variable capacitors are radio receivers, where these are used for frequency adjustment. Semi-variable or trim capacitors are miniature capacitors, with capacity ranging from several pF to several tens of pFs. These are used for fine tuning radio receivers, radio transmitters, oscillators, etc. Three trimmers, along with their symbol, are shown on the figure 2.5d.



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